Research notes and remaining questions

A guide to the current results and exploratory targets

21 October 2026

Abstract

This companion distinguishes proved results, conditional applications and questions recorded during the Alexander research project. The complete pseudo-visibility, species and quitting arguments have their own current papers. We give a self-contained finite-observation obstruction and explain its conditional application to a companion population construction. The remaining cards state the additional semantic or quantitative choices required before an exploratory suggestion becomes a theorem target. A recorded question is not a certification of present-day openness or novelty.

1. Where the full arguments are now presented

The project's results concern several different notions of learning, representation and comparison. Keeping them in separate full papers makes their assumptions easier to assess.

Pseudo-visibility. The full paper *Information and learning in pseudo-visibility environments* separates deterministic policy representation, private-query value, shared random seeds and learning dynamics. It gives exact one-step values and a restricted sample-average tabular convergence theorem. The included finite-horizon experiments are evidence about specified learners, rather than a universal wrapper-improvement theorem.

Species. The full paper *Specieslike clusters with several founders* combines founding cohorts and fixed founder bounds, proves coverage, compares infinite generator-equivalence classes, and gives the Borel complexity and binary non-learning results for IAP. The equivalence statement concerns infinite clusters. A theorem about deciding IAP does not by itself rule out every notion of convergence for species-valued outputs.

Quitting. The full paper *Quitting thresholds and reward normalization in universal intelligence* combines the ultrafilter boundary with pathwise and signed expected-value models. For the latter, its counterexample has quantifiers $\forall r\, \exists U\, \exists A, B$. The universal reference machine is chosen as part of the witness.

These are current arguments with their assumptions stated directly. This companion does not provide alternative versions of their proofs.

2. A finite-observation obstruction

Consider an infinite binary stream s. A learner sees successive finite prefixes and outputs a binary guess. Eventual correctness means that its guesses stabilize to the right answer on every input; the

learner need not be computable.

Proposition. Eventual periodicity has no universally correct learner

No finite-prefix learner is eventually correct on every stream about whether that stream is eventually periodic.

Proof. Let EP be the eventually periodic streams in Cantor space. It is countable: a finite initial segment and finite period specify a member. It is dense: every finite prefix can be continued periodically. Each singleton is nowhere dense, so EP is meager.

If the guesses g_n, based on length-n prefixes, converged correctly on every input, then

$$\mathrm{EP} = \bigcup_N \bigcap_{n \geq N} \{g_n = 1\}$$

would be F_σ, since each guess set is clopen. The same expression with zero would make its complement F_σ, hence EP would also be G_δ. But a dense G_δ subset of Cantor space is comeager. It cannot also be meager, by the Baire category theorem. This is a contradiction. □

The argument is topological; it does not use a limitation of computer running time. Any task continuously encoding this yes/no property inherits the obstruction.

The public companion *Biological unavoidability results* gives a population family P_s for which realizing every infinite binary sequence is equivalent to eventual periodicity of s. Under that construction and its finite-prefix observation interface, the proposition gives non-learning of its universality property. The construction itself is a separately cited input, not proved here.

Finite certification, one-sided detection and eventual two-sided correctness are different requirements. The property that a 1 occurs is detected at its first occurrence. An algorithm can start with no and switch to yes then. A prefix containing a 1 certifies the positive answer; an all-zero prefix cannot certify the negative one. Promises can change a learning problem as well: a stream promised to repeat a specified-length period after a known cutoff is determined by a sufficiently long prefix.

3. Pattern arithmetic: specify the representation first

Alexander's *Arithmetical algorithms for elementary patterns* (Archive for Mathematical Logic, 2015, DOI: 10.1007/s00153-014-0404-9), Section 6, proposes further algorithms for epsilon, Gamma and Veblen functions, and for second-order patterns.

A useful algorithmic target must fix the pattern class, coding of inputs, notational range, operation and correctness interpretation. For example, an epsilon-function algorithm would take a pattern denoting α, return one denoting ε_α, and establish correctness under the chosen pattern/ordinal correspondence. A two-argument Veblen algorithm needs an equally explicit domain and convention for $\varphi_\alpha(\beta)$.

These are recorded specifications, not claims that arbitrary pattern arithmetic is elementary, that small tests prove correctness, or that the literature has left each task open. A new selection requires checking the precise existing algorithm and current source version.

4. Variadic Apply: the function argument is part of the problem

Alexander's *The first-order syntax of variadic functions* (Notre Dame Journal of Formal Logic 54(1), 2013, arXiv:1105.4135), Section 7, motivates an Apply operation:

$$\mathrm{Apply}(G, n_1, \ldots, n_k) = G(n_1, \ldots, n_k).$$

Giving a sort for finite sequences does not by itself make the variable function G available.

An exact target must choose a function sort, a code domain with an evaluation relation, or an indexed operator family. It must say whether the functions are all intended functions or a specified represented collection. Its completeness claim must distinguish ordinary first-order models from any intended full function semantics. Conservativity also needs an explicit original language and class of reducts.

Until those choices are made, the card is a semantic design question rather than a proved completeness theorem.

5. Questions retained with their limits

Internodon membership and persistence

The internodon paper proves a clade-change bound and states a separate merge-tree height-stability theorem. Neither directly controls the co-membership of fixed organisms.

A possible next question is whether an additional persistence margin, together with an organism-label correspondence and the SD/test-organism rules, guarantees membership stability. The separating branch and margin must first be defined for that labelled construction. Ordinary diagram stability alone does not supply this implication. The saved all-pairs percentages do not establish it either.

Intelligence sign for a fixed symmetric machine

Alexander–Hutter's *Reward-punishment symmetric universal intelligence* (2021, DOI: 10.1007/978-3-030-93758-4_1), Section 5, raises sign computability. Its duality gives zero to reward-ignoring agents.

A recorded reduction strategy would run a reward-ignoring agent until a program halts, then switch to one of finitely many reward-sensitive rules. To decide halting from the strong sign, one needs a uniform non-vanishing statement: for every switch time and every reference machine under consideration, at least one switched rule has nonzero value. That statement is not supplied here. Adding the reward dual of a rule only negates its value and does not repair vanishing. The weak-sign variant has its own promise. Neither the quitting paper nor finite special-prior constructions settle this fixed-machine question.

Introspection outside the unstratified lemma

Entry 07 analyzes the unstratified introspection bundle. The stratified comparison concerns Lemma 65 and Corollary 71 of *Self-referential theories* (2020). A proposed extension must check where compactness, assigned validity and closure operate, and whether closure only moves upward in the ordinal index. Reusing the unstratified proof's wording does not establish that transfer. This card records the dependency rather than claiming a new minimality theorem.

Credit assignment and learning rates

The pseudo-visibility paper proves equal greedy limits only for its finite tabular model with sample averages, unique maximizers and explicit sampling/noise assumptions. Extending that result to all algorithms that update erased parameters from erased rewards is a broader question. A bound on the gap, or on constant versus context-linear sample complexity, needs specified rewards, learner, features, accuracy and confidence. The existing experiment does not pay those requirements.

Representation and music complexity

Alexander's playful *Legg–Hutter universal intelligence implies classical music is better than pop music for intellectual training* (The Reasoner, 2019) invites a representation-level comparison. A score, a recording and a performance conditional on a score are different coding objects. A meaningful comparison should hold that level fixed and state the compressor, transcription and conditioning information.

This is an exploratory empirical suggestion, not a proved genre ranking or a conclusion about Kolmogorov complexity from file sizes. No experiment for it is included.

6. How to use the cards

Each card separates a specified result from the missing semantic, quantitative or source-status input. The purpose is to make a next question assessable. A saved conjecture does not establish present-day openness, and a finite control does not replace an infinite proof.

The full papers linked in Section 1 provide their own sources, assumptions and reproducibility guides. The remaining cards record possibilities without claiming a literature survey, research priority or new resolution.